

1 (SET A)

$$HSV = (270^\circ, 40\%, 90\%) = (270^\circ, 0.4, 0.9)$$

According to Fig 1,

$C_{max} \rightarrow \text{Blue}$

$C_{min} \rightarrow \text{Green}$

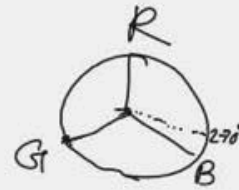


Fig 1

$$\text{Now, } V = C_{max} = 0.9 \text{ (B)}$$

$$S = \frac{C_{max} - C_{min}}{C_{max}}$$

$$\Rightarrow 0.4 = \frac{0.9 - C_{min}}{0.9}$$

$$\Rightarrow C_{min} = 0.54 \text{ (G)}$$

$$\text{Again, } H = \left(4 + \frac{R - G}{\text{diff}}\right) \times 60^\circ$$

$$\Rightarrow 270^\circ = \left(4 + \frac{R - 0.54}{0.36}\right) \times 60^\circ$$

$$\Rightarrow R - 0.54 = \left(\frac{270^\circ}{60^\circ} - 4\right) \times 0.36$$

$$\begin{aligned} \Rightarrow R &= 0.5 \times 0.36 + 0.54 \\ &= 0.72 \end{aligned}$$

$$\therefore RGB = (0.72, 0.54, 0.9)$$

Ans.

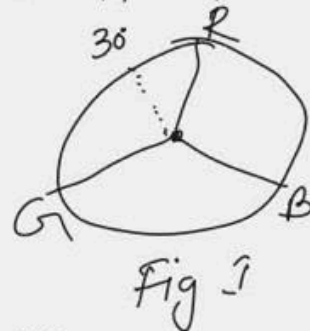
1 (SET B)

$$HSV = (30^\circ, 40\%, 90\%) = (30^\circ, 0.4, 0.9)$$

According to Fig 1,

$C_{max} \rightarrow \text{Red}$

$C_{min} \rightarrow \text{Blue}$



$$\text{Now, } V = C_{max} = 0.9 \text{ (R)}$$

$$S = \frac{C_{max} - C_{min}}{C_{max}}$$

$$\Rightarrow 0.4 = \frac{0.9 - C_{min}}{0.9}$$

$$\Rightarrow C_{min} = 0.54 \text{ (B)}$$

$$\text{Again, } H = \left(0 + \frac{G - B}{\text{diff}}\right) \times 60^\circ$$

$$\Rightarrow 30^\circ = \left(0 + \frac{G - 0.54}{0.36}\right) \times 60^\circ$$

$$\Rightarrow G - 0.54 = \left(\frac{30^\circ}{60^\circ} - 0\right) \times 0.36$$

$$\begin{aligned} \Rightarrow G &= 0.5 \times 0.36 + 0.54 \\ &= 0.72 \end{aligned}$$

$$\therefore RGB = (0.9, 0.72, 0.54)$$

Ans.

1 (SET C)

$$HSV = (150^\circ, 40\%, 90\%) = (150^\circ, 0.4, 0.9)$$

According to Fig 1,

$C_{max} \rightarrow \text{Green}$

$C_{min} \rightarrow \text{Red}$

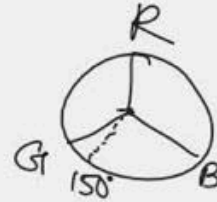


Fig 1

$$\text{Now, } V = C_{max} = 0.9 \text{ (G)}$$

$$S = \frac{C_{max} - C_{min}}{C_{max}}$$

$$\Rightarrow 0.4 = \frac{0.9 - C_{min}}{0.9}$$

$$\Rightarrow C_{min} = 0.54 \text{ (R)}$$

$$\text{Again, } H = \left(2 + \frac{B - R}{\text{diff}}\right) \times 60^\circ$$

$$\Rightarrow 150^\circ = \left(2 + \frac{B - 0.54}{0.36}\right) \times 60^\circ$$

$$\Rightarrow B - 0.54 = \left(\frac{150^\circ}{60^\circ} - 2\right) \times 0.36$$

$$\begin{aligned} \Rightarrow B &= 0.5 \times 0.36 + 0.54 \\ &= 0.72 \end{aligned}$$

$$\therefore RGB = (0.54, 0.9, 0.72)$$

Ans.

2 (SET A)

$$CMY = (255, 85, 170) = \left(1, \frac{1}{3}, \frac{2}{3}\right)$$

$$RGB = \left(0, \frac{2}{3}, \frac{1}{3}\right)$$

$$\begin{array}{l|l} C_{\max} = \frac{2}{3} \text{ (Green)} & \text{diff} = C_{\max} - C_{\min} \\ C_{\min} = 0 \text{ (Red)} & = \frac{2}{3} - 0 \\ V = C_{\max} = \frac{2}{3} & = \frac{2}{3} \end{array}$$

$$S = \frac{\text{diff}}{C_{\max}} = \frac{\frac{2}{3}}{\frac{2}{3}} = 1$$

$$\begin{aligned} H &= \left(2 + \frac{B - R}{\text{diff}}\right) \times 60^\circ = \left(2 + \frac{\frac{1}{3} - 0}{\frac{2}{3}}\right) \times 60^\circ \\ &= 150^\circ \end{aligned}$$

$$HSV = \left(150^\circ, 1, \frac{2}{3}\right) \quad (\text{Ans})$$

2 (SET B)

$$CMY = (85, 170, 255) = \left(\frac{1}{3}, \frac{2}{3}, 1 \right)$$

$$RGB = \left(\frac{2}{3}, \frac{1}{3}, 0 \right)$$

$$\begin{array}{l|l} c_{\max} = \frac{2}{3} & (\text{Red}) \\ c_{\min} = 0 & (\text{Blue}) \end{array} \quad \left| \begin{array}{l} \text{diff} = c_{\max} - c_{\min} \\ = \frac{2}{3} - 0 \\ = \frac{2}{3} \end{array} \right.$$

$$S = \frac{\text{diff}}{c_{\max}} = \frac{\frac{2}{3}}{\frac{2}{3}} = 1$$

$$H = \left(0 + \frac{G-B}{\text{diff}} \right) \times 60^\circ = \frac{\frac{1}{3} - 0}{\frac{2}{3}} \times 60^\circ = 30^\circ$$

$$HSV = \left(30^\circ, 1, \frac{2}{3} \right) \quad (\text{Ans})$$

2 (SET C)

$$CMY = (85, 255, 170) = \left(\frac{1}{3}, 1, \frac{2}{3}\right)$$

$$RGB = \left(\frac{2}{3}, 0, \frac{1}{3}\right)$$

$$\begin{array}{l|l} C_{\max} = \frac{2}{3} \text{ (Red)} & \text{Diff} = C_{\max} - G_{\min} \\ C_{\min} = 0 \text{ (Green)} & = \frac{2}{3} - 0 \\ V = C_{\max} = \frac{2}{3} & = \frac{2}{3} \end{array}$$

$$S = \frac{\text{Diff}}{C_{\max}} = \frac{\frac{2}{3}}{\frac{2}{3}} = 1$$

$$\begin{aligned} H &= \left(0 + \frac{G - B}{\text{Diff}}\right) \times 60^\circ = \left(\frac{0 - \frac{1}{3}}{\frac{2}{3}}\right) \times 60^\circ \\ &= -30^\circ = 330^\circ \end{aligned}$$

$$HSV = \left(330^\circ, 1, \frac{2}{3}\right) \quad (\text{Ans})$$

3 & 4

③

Ⓐ The point where parallel lines meet in perspective projection

Ⓑ Infinite

Ⓒ Perspective projection

Ⓓ perspective : projected image becomes smaller

Parallel : Doesn't change

④ Ⓐ Perspective . Because it is more realistic .

Ⓑ Parallel projection .

Ⓒ $(-150, 0, 40)$

